Four-Dimensional Solids Whose Every Cell Is a Golden Cut



Each regular solid meets its dual in a six-dimensional product, and the tesseract meets the 16-cell in eight: there every cell is a golden cut, Icosaxes beside the Lesser Golden Cut Tetrahedron and Unborn Axes beside the Greater; OG and the polar Dharma cycle are stations of the dodecahedron's sweep

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Paper #91 · version 1 · 4 November 2026 · DICTO DOI: 10.5281/zenodo.23931389 (concept) · Licence CC BY-SA 4.0 · Scripts and exact checks: studies/2026-11-03-dual-pairs-dimensions (research branch)
The proposal to follow the Platonic solids and their duals through the dimensions, the follow-up proposals (OG and the dodecahedron–icosahedron pair, new golden cycles, hyper-lattice cycles, cycles of the rhombic dodecahedron and truncated octahedron, the golden cut rhombic dodecahedra as four-dimensional slices), the golden cuts, OG, the polar Dharma cycle the names of the solids, among them Icosaxes, Unborn Axes and the Lesser and Greater Golden OG, and the names DICTO All-Golden Polytopes and DICTO Golden Tight Cycles are DICTO's. Computation, proofs, exact checks and drafting with Claude (Anthropic); an independent review by Fable (Anthropic) checked the main theorem.

Version 2 corrects the notes on methods: the names of our computers and private file paths are removed, with small wording fixes. No result changes.

> *Put a solid and its dual side by side in a space of twice the dimension and cut that space along a tilted flat. What you see in the cut is the solid trimmed by its dual, a solid with its corners cut, and the depth of the cut is set by the tilt. As the tilt turns, the cut passes through the whole family, from the dual to the solid itself, and every golden cut is a station on the way. Do the same with the four-dimensional cube and its dual, the 16-cell, and two of the cuts are four-dimensional solids whose every cell is a golden cut: Icosaxes beside the Lesser Golden Cut Tetrahedron, and Unborn Axes beside the Greater. The dodecahedron's pair carries DICTO's Ogdoecontahedron and the whole polar Dharma cycle as stations of the same sweep, and new cycles close by the golden ratio.*

Abstract. DICTO proposed following each Platonic solid with its dual through the dimensions. For a regular polyhedron P and its polar P*, the flats $y = x/\mu$ of the product P × P* cut out $S(\mu) = P \cap \mu P^*$, and we prove that every corner cut of P or P* is such a section, so all ten golden cuts of #88 lie on the three sweeps D·I, C·O and T·T. Our main theorem is four-dimensional: in the tesseract K trimmed by its polar 16-cell, $K \cap \mu K^*$, every cell is a golden cut exactly when $\mu = \varphi^2$ or $\mu = 2 + 1/\varphi^2$; the 24 cells are then 8 Icosaxes and 16 Lesser Golden Cut Tetrahedra, or 8 Unborn Axes and 16 Greater Golden Cut Tetrahedra, because the golden heights of the cube cells and of the tetrahedral cells coincide; these two are DICTO All-Golden Polytopes, four-dimensional solids all of whose cells are golden cuts. In the 600-cell × 120-cell family, by a linear cell rule checked exactly at three points, the bitruncated stage gives DICTO All-Golden Polytopes at $\mu = 2 - \sqrt{5}/2$ and $(19 + 9\sqrt{5})/44$. In three dimensions the D·I sweep lies entirely in $\mathbb{Q}(\sqrt{5})$ but no lattice carries its symmetry; the tesseract's slices along a long diagonal are the T·T sweep in a Möbius coordinate; and the DICTO Ogdoecontahedron is exactly $OG = H(\mu_{TI}) \cap \varphi^2 D^*$, the hull of the dodecahedron and the scaled icosahedron at the truncated-icosahedron

level cut by icosahedron planes at the rectification level, so that the four middle stations of the polar Dharma cycle are sections and hulls of the six-dimensional product, unscaled and touching; at the golden levels √5 and (13 + 6√5)/11 the same construction gives DICTO's Lesser and Greater Golden OG, with OG's combinatorics. Tight cycles that close at a power of φ, the DICTO Golden Tight Cycles, occur at φ for D·I and C·O and at φ^6 for T·T; slices of the 24-cell honeycomb give Icosaxes + Unborn Axes at $1/\varphi^2$ and Kelvin's truncated octahedra at 1/2. On the cube side √2 selects the silver ratio throughout. All results are exact in Q(√2, √5) except where stated.

1. Introduction

#88 proved that each Platonic solid has exactly two golden cuts [1]; #89 found all five golden pairs as slices of four-dimensional solids [2]; and #90 proved that the Golden Cut Cubes lie outside every golden structure and placed the cube–octahedron pair, as DICTO proposed, in the six-dimensional product of a cube and an octahedron, where the lattice selects Icosaxes at the Fibonacci slope [3]. DICTO then proposed following the other Platonic solids with their duals through the dimensions in the same way, and in follow-ups proposed connecting OG [4] with the dodecahedron–icosahedron pair, new golden cycles, hyper-lattice cycles, cycles through the rhombic dodecahedron and the truncated octahedron, and the golden cut rhombic dodecahedra of #89 as four-dimensional slices.

The strongest result is four-dimensional and comes first (Section 3): the tesseract trimmed by its dual has two members all of whose cells are golden cuts. Section 4 treats the three sweeps in three dimensions, Section 5 places OG, the Golden OGs and the polar Dharma cycle in the dodecahedron's sweep, Section 6 the other four-dimensional pairs, Sections 7 and 8 the cycles and the honeycomb slices, and Section 9 collects the pattern: golden where the pair lives in Q(√5), silver where the square's √2 enters.

Names. The golden cuts, Icosaxes (Greater Golden Cut Octahedron) and Unborn Axes (Lesser Golden Cut Octahedron) are DICTO's names from #88, with Greater for the deeper cut; GCD, GCI, GCC and GCT abbreviate Golden Cut Dodecahedron, Icosahedron, Cube and Tetrahedron. OG is DICTO's Ogdoecontahedron [4]. DICTO named the two golden relatives of OG found here the **Lesser Golden OG** (at μ = √5) and the **Greater Golden OG** (at μ = (13 + 6√5)/11), together the Golden OGs. The polar Dharma cycle is DICTO's [5, 6].

2. The sweep of a dual pair

Let P be a regular polyhedron with circumradius R, midradius ρ, inradius r and edge e, written as $P = \{n \cdot x \le 1\}$ with n over its face poles, and let $P^* = \mathrm{conv}(n)$ be its polar. In the six-dimensional product $P \times P^*$, the flats $F_\mu = \{(x, x/\mu)\}$ carry the diagonal symmetry (Schur's lemma, as in #90), and the section is $S(\mu) = P \cap \mu P^*$. Its polar is the hull: $\mathrm{polar}(S(\mu)) = H(\mu)/\mu$ with $H(\mu) = \mathrm{conv}(P \cup \mu P^*)$. We write $\nu = \mu/r^2$, so that the dual end is $\nu = 1$. The same definitions hold for a regular four-dimensional solid in the eight-dimensional product.

The cut of #88: an n-gon face cut at edge fraction x becomes a 2n-gon with corner sides $c_n x$, $c_n = 2\cos(\pi/n)$, and original sides $1 - 2x$; it is golden at $x_G = \varphi/(c_n + 2\varphi)$ (Greater) and $x_L = 1/(\varphi c_n + 2)$ (Lesser) [1]. For n = 3 the face is a truncated triangle.

3. Main theorem: the sweep holds every golden cut, and the DICTO All-Golden Polytopes

> **Main Theorem.** (a) *(Lemma 1.)* For a regular polyhedron P, S(μ) runs from r^2P^* ($\nu = 1$) through the rectified solid ($\mu = \rho^2$) to P ($\mu = R^2$). Cutting each corner of P at edge fraction x, $0 < x \le ½$ (that is, $\rho^2 \le \mu \le R^2$), gives S(μ) with $\mu = R^2 - xe^2/2$; cutting each corner of P* at fraction x, $0 < x \le ½$, gives S(μ) with $1/\mu = R^{*2} - xe^{*2}/2$. So every corner cut lies on the sweep, and the ten golden cuts of #88 are stations of the three sweeps D·I, C·O and T·T.
> (b) Let $K = [-1, 1]^4$ be the tesseract and $K^* = \{|x|_1 \le 1\}$ its polar 16-cell. For $1 < \mu < 4$, $S(\mu) = K \cap \mu K^*$ has 24 three-dimensional cells, 8 on the facets of K and 16 on the facets of μK^*; the integer points $\mu = 2$ and $\mu = 3$ are the 24-cell and the rectified tesseract, which are not golden. Every cell is a golden cut exactly when $\mu = \varphi^2$ or $\mu = 2 + 1/\varphi^2$. At $\mu = \varphi^2 = 2 + 1/\varphi$ the cells are **8 Icosaxes and 16 Lesser Golden Cut Tetrahedra**; at $\mu = 2 + 1/\varphi^2$ they are **8 Unborn Axes and 16 Greater Golden Cut Tetrahedra**. These two polytopes are **DICTO All-Golden Polytopes**, DICTO's name for the four-dimensional polytopes all of whose cells are golden cuts.

Proof. (a) The plane $v \cdot y = \mu$ is perpendicular to the corner axis v of P and meets the edge from v to a neighbour w at $v + x(w - v)$ exactly when $\mu = v \cdot v - x(v \cdot v - v \cdot w) = R^2 - xe^2/2$; the dual side is the same statement for P*, read through $S(\mu) = \mu(P^* \cap \mu^{-1}P)$. At $x = 0$, 1/2 and the dual end this gives P, the rectified solid and r^2P^*.
(b) A facet of S(μ) lies on a facet of K or of μK^*. On the facet $x_4 = 1$ of K the cell is the cube $[-1, 1]^3$ cut by $|x_1| + |x_2| + |x_3| \le \mu - 1$, that is $C \cap \lambda O$ with $\lambda = \mu - 1$: the section of cube × octahedron of #90 at λ. On the facet $s \cdot x = \mu$ of μK^* ($s \in \{\pm 1\}^4$) every point has $s_i x_i \ge 0$, since $|x|_1 \le \mu = s \cdot x$; so the cell is $\{0 \le s_i x_i \le 1, \Sigma\, s_i x_i = \mu\}$, a slice of the unit tesseract across its long diagonal. With $y_i = 2s_i x_i - 1$ it is the slice $y_1 + y_2 + y_3 + y_4 = 2h$ of $[-1, 1]^4$ with $h = \mu - 2$. For every μ in (1, 4), the integers 2 and 3 included, both kinds of cell are three-dimensional, so there are 8 + 16 = 24 cells.
For $1 < \mu < 2$ the cube cells are regular octahedra λO ($\lambda < 1$), and for $3 < \mu < 4$ the tetrahedral cells are regular tetrahedra ($h > 1$, a corner of the slice); neither is a golden cut, since a golden cut has $x > 0$ (#88, Theorem 2: both roots lie in (0, ½)) and these cells are the uncut solids. For $2 < \mu < 3$ the cube cells are octahedra cut at their corners ($1 < \lambda < 2$, corners the signed permutations of $(1, \lambda - 1, 0)$); the plane $x_1 = 1$ meets the edge from (λ, 0, 0) to (0, λ, 0) at fraction $x = (\lambda - 1)/\lambda$, so by (a) for the octahedron ($R = 1$, $e = \sqrt{2}$: $1/\lambda = 1 - x$) and #88 they are golden exactly when $x = 1/\varphi^2$ (Icosaxes, $\lambda = \varphi$) and $x = 1/(\varphi\sqrt{5})$ (Unborn Axes, $\lambda = 3 - \varphi$) [3, Table 2]. The tetrahedral cells, by #89's Main Theorem (a), are regular tetrahedra cut at $x = (1 - h)/(2 - h)$, golden exactly at $h = 1/\varphi^2$ (Greater) and $h = 1/\varphi$ (Lesser) [2]. Now $\lambda = \varphi$ means $\mu = 1 + \varphi = \varphi^2 = 2 + 1/\varphi$, that is $h = 1/\varphi$; and $\lambda = 3 - \varphi$ means $\mu = 4 - \varphi = 2 + 1/\varphi^2$, that is $h = 1/\varphi^2$. The two golden sets coincide, which with the preceding gives (b). Both stations were also built exactly and matched corner to corner (s5). □

The coincidence has a local reason. A hexagon of a cube cell is shared with a tetrahedral cell, and it is a truncated triangle in both. Its sides alternate between the cut sides of the one cell and the original sides of the other, so a ratio φ in the one is $1/\varphi$ in the other: the hexagons of Icosaxes (Greater) are those of the Lesser Golden Cut Tetrahedron, and the hexagons of Unborn Axes (Lesser) those of the Greater (s5, exact: the six corners of the shared hexagon and its sides $\sqrt{2} \cdot h$ and $\sqrt{2} \cdot (1 - h)$ are computed at both stations). At $\mu = 5/2$, halfway between, both cells are Archimedean, truncated octahedra and truncated tetrahedra: the uniform

bitruncated tesseract. The tesseract × 16-cell sweep thus carries the C·O sweep (cube cells, $\lambda = \mu - 1$) and the T·T sweep (tetrahedral cells, linear in the four-dimensional height) at once.

In the lower layer $1 < \mu < 2$, the truncated-16-cell stage, the tetrahedral cells are golden at $\mu = 3 - \varphi$ (Lesser) and $\mu = \varphi$ (Greater), beside regular octahedra: every cell is golden or regular. In the upper layer $3 < \mu < 4$, the truncated-tesseract stage, the cube cells are the Golden Cut Cubes at $\mu = 1 + \mu_G$ and $1 + \mu_L$, of degree 4 (#90), beside regular tetrahedra (s5, exact).

4. The three sweeps in three dimensions

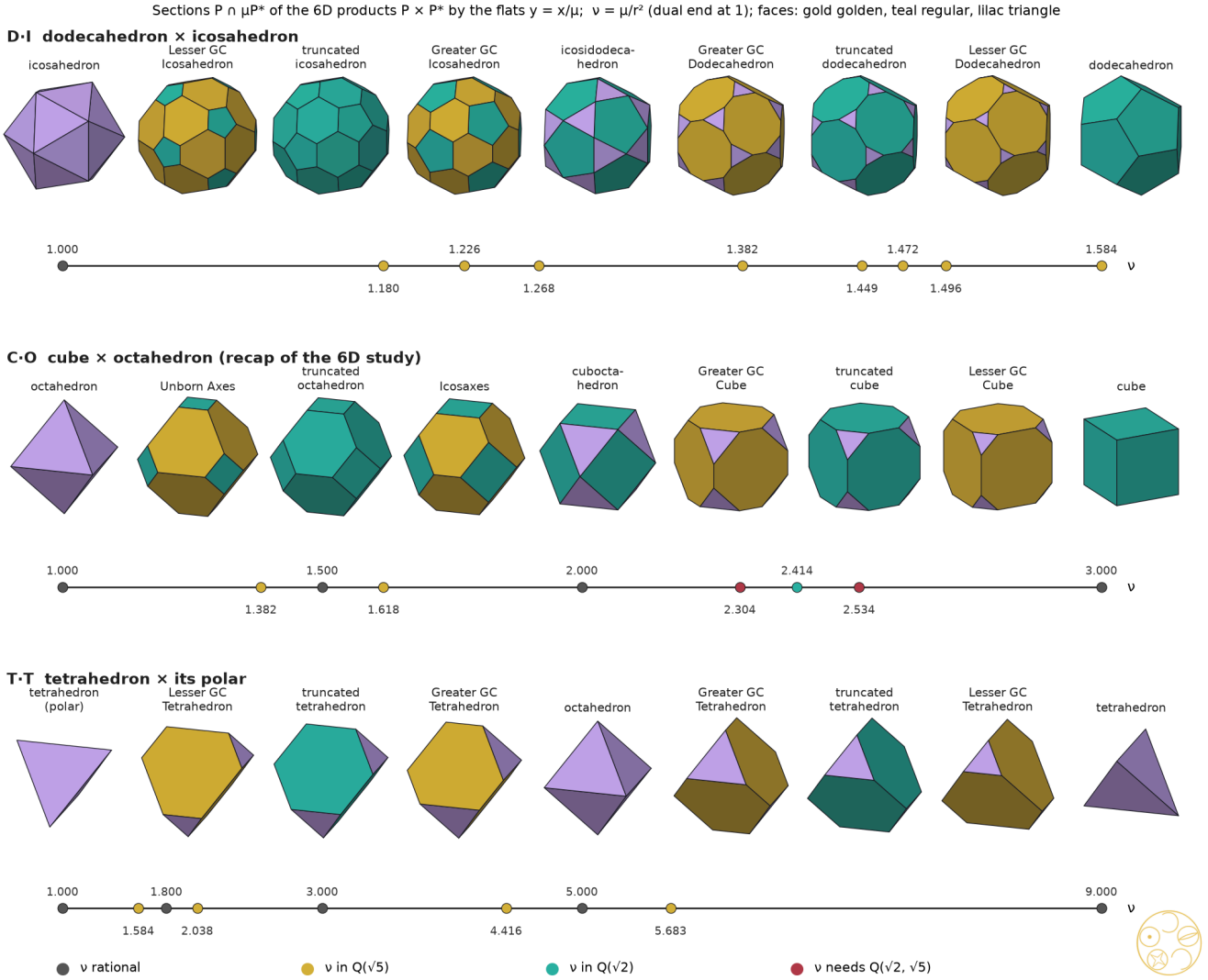


Figure 1. The three sweeps $S(\mu) = P \cap \mu P^*$: dodecahedron × icosahedron, cube × octahedron and tetrahedron × its polar, from the dual end (left) to the solid (right). Exact sections drawn in floating point; faces gold golden, teal regular, lilac triangles. Below each strip, the stations on the ν-line ($\nu = \mu/r^2$, dual end at 1), coloured by the field of ν: grey rational, gold Q(√5), teal Q(√2), red Q(√2, √5).

Dodecahedron × icosahedron (D·I). Take D with corners $(\pm 1, \pm 1, \pm 1)$ and $(0, \pm 1/\varphi, \pm\varphi)$ cyclic, the cube's own frame, with $R^2 = 3$ for the cube, the dodecahedron and the tetrahedron alike. Here $r^2 = 1 + 2/\sqrt{5}$, $\rho^2 = \varphi^2$ and $R^2 = 3$. All nine stations were built exactly and matched corner to corner with the cut solids (s1, 20 checks).

station	μ ($R^2 = 3$)	ν = μ/r^2	field norm of ν
icosahedron (dual end)	$1 + 2/\sqrt{5}$	1	1
Lesser Golden Cut Icosahedron	$\sqrt{5}$	$5\sqrt{5} - 10 = 5/\varphi^3$	−25
truncated icosahedron	$(24 + 9\sqrt{5})/19$	$(30 - 3\sqrt{5})/19$	45/19
Greater Golden Cut Icosahedron	$(13 + 6\sqrt{5})/11$	$(5 + 4\sqrt{5})/11$	−5/11
icosidodecahedron	φ^2	$3 - \varphi$	5
Greater Golden Cut Dodecahedron	$2 + \sqrt{5}/3$	$(20 - 7\sqrt{5})/3$	155/9
truncated dodecahedron	$1 + 4/\sqrt{5}$	$2\sqrt{5} - 3$	−11
Lesser Golden Cut Dodecahedron	$(20 + 5\sqrt{5})/11$	$(50 - 15\sqrt{5})/11$	125/11
dodecahedron	3	$15 - 6\sqrt{5} = (R/r)^2$	45

Table 1. The D·I sweep (s1, exact).

Theorem 2 (the D·I sweep). All four golden cuts of the dodecahedron and icosahedron lie on the D·I sweep, and every μ and ν is in $\mathbb{Q}(\sqrt{5})$: there is no √2 obstruction. Two values are natural in the cube's frame: the rectified stage at $\mu = \varphi^2$ and the Lesser Golden Cut Icosahedron at $\mu = \sqrt{5}$. No golden ν is a power of φ, since their field norms are not ±1. Across all three pairs, the only golden station whose ν is a power of φ is Icosaxes ($\nu = \varphi$).

The values follow from Lemma 1 with $R^2 = 3$ and $e = 2/\varphi$ (for example the Greater Golden Cut Dodecahedron, $x = 1/3$: $\mu = 3 - (1/3)(2/\varphi)^2/2 = 2 + \sqrt{5}/3$), and were checked exactly (s1). Two coincidences recur across pairs, exact but with no structural reason found: the D·I rectification $\nu = 3 - \varphi$ equals the μ of Unborn Axes, and the truncated dodecahedron's $\nu = 2\sqrt{5} - 3$ equals the μ of the Greater Golden Cut Tetrahedron.

Theorem 3 (no lattice for D·I). No lattice in R^6 is invariant under the icosahedral group acting diagonally on D × D*, so the cut-equals-shadow test of #90 cannot be posed for this pair. In the icosahedral $Z^6 = V \oplus V'$ the only invariant 3-flats are V and V′, so there is no one-parameter family of symmetric flats.

Proof. An invariant lattice makes the representation rational. A five-fold rotation has trace $1 + 2\cos 72° = \varphi$ on V (checked on the exact matrix), hence 2φ on $V \oplus V$, which is irrational. Any rational representation containing V also contains its Galois twin V′ (trace $1 - \varphi$), and V and V′ are not isomorphic. □ What Z^6 does pick out is exact: the central sections of the Voronoi cells of Z^6, D6* and D6 are the dodecahedron, icosidodecahedron and rhombic triacontahedron; the shortest Z^6 vectors cast an icosahedron, the D6 roots two icosidodecahedra with radii in ratio φ, and the vectors $(\pm 1/2)^6$ only dodecahedra and icosahedra (s2, 9 checks). These are the ends and the rectified stage only, with no golden cut among them (Al-Siyabi, Koca and Koca derived icosahedral polyhedra from D6 [11]).

Cube × octahedron (C·O). #90's sweep is reproduced exactly (s1, 22 checks): octahedron 1, Unborn Axes $3 - \varphi$, truncated octahedron 3/2, Icosaxes φ, cuboctahedron 2, Greater Golden Cut Cube ≈ 2.304114, truncated cube $1 + \sqrt{2}$, Lesser Golden Cut Cube ≈ 2.533609, cube 3. Here $r^2 = 1$, so $\nu = \mu$. Its polar side is #54's Theorem B [16]: for $1 \le \mu \le 2$ the hull $H(\mu) = \mathrm{conv}(\text{cube} \cup \mu\cdot\text{octahedron})$ is the cube with a square pyramid of height $\mu - 1$ (in half cube edges) on each face, so the polar of Icosaxes ($\mu = \varphi$) has pyramids of height $1/\varphi$ and that of Unborn Axes ($\mu = 3 - \varphi$) height $1/\varphi^2$; along #48's family [8] the heights $x/(1 + x)$ and $1/(1 + x)$, at μ and $3 - \mu$, add up to 1, the rhombic dodecahedron H(2).

Tetrahedron × tetrahedron (T·T). With T inscribed in the cube $[-1, 1]^3$, the polar is −T and $r^2 = 1/3$. The sweep runs −T ($\mu = 1/3$), Lesser GCT $5 - 2\sqrt{5}$, truncated tetrahedron 3/5,

Greater GCT $(3 + 2\sqrt{5})/11$, octahedron 1, Greater GCT $2\sqrt{5} - 3$, truncated tetrahedron 5/3, Lesser GCT $1 + 2/\sqrt{5}$, and T at 3; the self-duality $\mu \leftrightarrow 1/\mu$ is exact (s1, 20 checks).

> **Theorem 4 (the tesseract's slices are the T·T sweep).** The section of $[-1, 1]^4$ by $x_1 + x_2 + x_3 + x_4 = 2h$ $(0 < h < 1)$ is similar to $T \cap \mu(-T)$ with $\mu = (2 + h)/(2 - h)$. The ratio of corner side to remaining side is $(1 - h)/h$, so the Greater and Lesser Golden Cut Tetrahedra fall at #89's heights $h = 1/\varphi^2$ ($\mu = 2\sqrt{5} - 3$) and $h = 1/\varphi$ ($\mu = 1 + 2/\sqrt{5}$), and the truncated tetrahedron at $h = 1/2$ ($\mu = 5/3$).

Proof. In the hyperplane the constraints $x_i \le 1$ and $x_i \ge -1$ bound two opposite regular tetrahedra, of sizes proportional to $2 - h$ and $2 + h$, so the section is $T \cap \mu(-T)$ with $\mu = (2 + h)/(2 - h)$. Checked exactly at the three heights in a rational orthonormal frame (s3). □ The lattice $Z^3 \oplus Z^3$ carries T_d, unlike D·I, but its test does not discriminate: the vectors $(0, 1, 1 \mid 2, -1, -1)$ and $(-2, 1, 1 \mid 0, -1, -1)$, of norm 8, cast the section's corners at every μ, and the shortest isolated selectors of the golden-cut tetrahedra have norm 34 (Lesser) and 46 (Greater), minima in the box $\{-6, \ldots, 6\}^6$ confirmed exactly, against norm 3 for $\mu = 2$ and 11 for the Archimedean 5/3. No golden selection on this side.

5. OG and the Golden OGs in the polar Dharma cycle

DICTO proposed connecting OG with the interactions of the dodecahedron and icosahedron at interdimensional points. The D·I pair has two one-parameter families: the sections $S(\mu)$ (Archimedean side) and their polars, the hulls $H(\mu) = \mathrm{conv}(D \cup \mu D^*)$ (Catalan side: pentakis dodecahedron at μ_{TI}, rhombic triacontahedron at φ^2, triakis icosahedron at μ_{TD}). Cutting a hull once more by icosahedron planes gives a two-parameter family $Q(\mu, \lambda) = H(\mu) \cap \lambda D^*$.

> **Theorem 5 (OG in the sweep).** The DICTO Ogdoecontahedron is exactly $OG = Q(\mu_{TI}, \varphi^2) = H(\mu_{TI}) \cap \varphi^2 D^*$, with $\mu_{TI} = (24 + 9\sqrt{5})/19$: the pentakis dodecahedron (the polar of the truncated icosahedron) cut by the icosahedron planes at the rectification level, the same planes that cut D down to the icosidodecahedron. The stations of DICTO's polar Dharma cycle [5, 6] are exactly
>
> $$(3/\varphi^3)D \subset S(\varphi^2) \subset Q(\mu_{TI}, \varphi^2) \subset H(\mu_{TI}) \subset H(3) = 3D^* \subset (3/\varphi^2)H(\varphi^2) \subset 3D,$$
>
> and the four middle stations are sections and hulls of the six-dimensional product themselves, unscaled, each touching the next (tight factor 1). At $\mu = \sqrt{5}$ and $\mu = (13 + 6\sqrt{5})/11$, the levels of the Lesser and Greater Golden Cut Icosahedra, $Q(\mu, \varphi^2)$ are DICTO's **Lesser and Greater Golden OG**, with the combinatorics of OG: 102 corners, 20 regular hexagons and 60 kites.

Proof (exact computation). $H(\mu_{TI})$ has 32 corners and 60 triangles (the pentakis dodecahedron); $Q(\mu_{TI}, \varphi^2)$ has 102 corners and 80 faces, 20 regular hexagons and 60 kites, and equals the polar-Dharma chain's OG corner for corner; each station of the chain equals the stated section or hull up to one common scale (s4, 21 checks); the Golden OGs were built the same way (s4c, 2 + 2 checks). □

The two end links are loose, with tight factors $(-99 + 45\sqrt{5})/2$ and $5 - 2\sqrt{5}$. The closure is φ^3 as given, and $225/2 - 99\sqrt{5}/2 \approx 1.814635$ when every link is tightened, matching the polar-Dharma study's tight nest [6]. So the cycle is a path through the sweep at only two parameter

values: the rectification level φ^2, a golden unit, and μ_{TI}, which is not golden (field norm 9/19). The closure φ^3 comes from the two loose ends, not from the sweep.

Lifts. In the module M cast by Z^6 (the integer span of (0, ±1, ±φ) cyclic), the smallest integer scale that lifts OG's corners into M is 228 in our frame and 342 = 2 × 171 in the chain frame, twice the bridge study's 171 [7], a factor that presumably comes from a different module convention (not re-derived). The prime 19 comes from μ_{TI}: the truncated icosahedron and the pentakis dodecahedron need 38, the dodecahedron and the Lesser Golden Cut Icosahedron 2, the Lesser Golden OG only 4, and the Greater Golden OG 220 (s4b, s4c). The sweep explains the 171 (it comes from the Archimedean depth, of field norm 9/19) without removing it; of the three, the Lesser Golden OG lifts most simply.

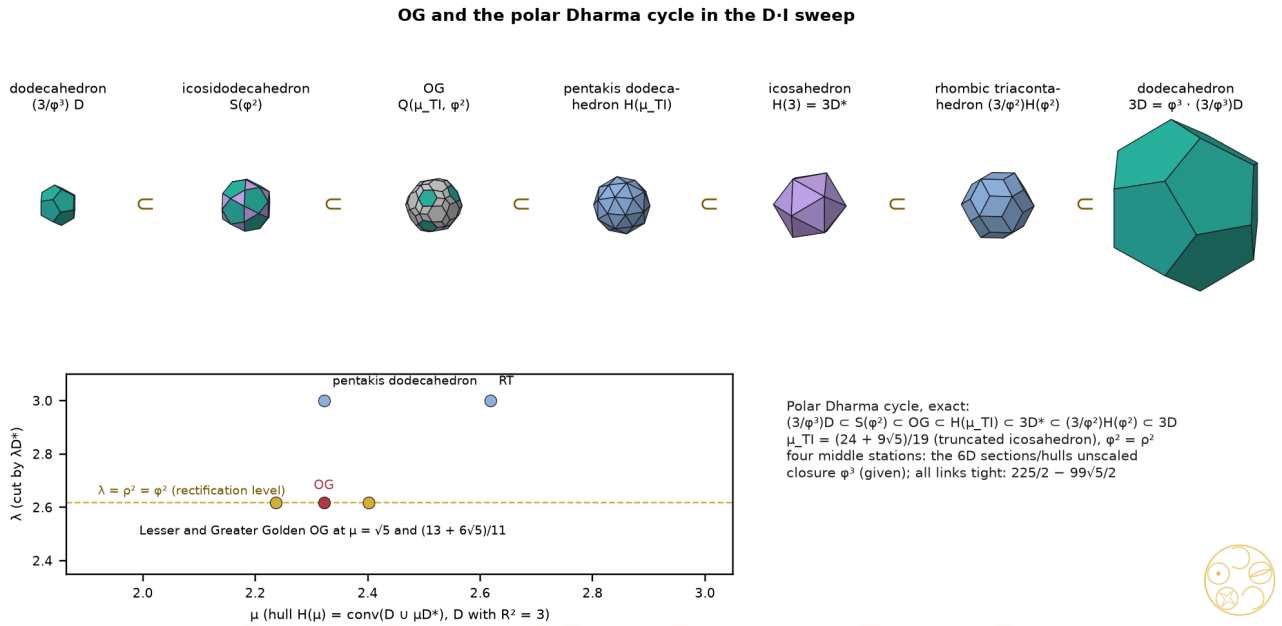


Figure 2. Above, the polar Dharma cycle as stations of the D·I families, drawn to scale: (3/φ³)D, the icosidodecahedron S(φ²), OG = Q(μ_{TI}, φ²), the pentakis dodecahedron H(μ_{TI}), the icosahedron H(3) = 3D*, the rhombic triacontahedron (3/φ²)H(φ²) and 3D = φ³·(3/φ³)D. Below, the (μ, λ) map: OG (red) and the Lesser and Greater Golden OG (gold, at μ = √5 and (13 + 6√5)/11) on the rectification level λ = φ², with the pentakis dodecahedron and the rhombic triacontahedron at λ = 3.

6. Four-dimensional duals in eight dimensions

For a regular four-dimensional solid P and its polar, the flats y = x/μ of P × P* ⊂ R^8 give P ∩ μP* again. As μ grows the section passes through P*, truncated P*, rectified P*, the bitruncated stage, rectified P, truncated P and P; the stage changes at the squared distances of the centres of the k-faces.

> **Lemma 6 (cells; floating point beyond three exact checks).** A cell of P ∩ μP* on a facet of P is that facet's own three-dimensional sweep, shifted linearly in μ (for the tesseract $\mu_3 = \mu - 1$, Section 3); cells on the facets of μP* follow the same rule in 1/μ. For the tesseract and the 24-cell this was checked exactly (rational frames, corner matching). For the 600-cell × 120-cell family it was checked exactly at three stations and in floating point everywhere; it is not proved in general.

pair	golden or notable μ (exact)	what the cells are
tesseract × 16-cell (μ ∈ [1, 4])	$3 - \varphi$, φ; $2 + 1/\varphi^2$, φ^2; $1 + \mu_G$, $1 + \mu_L$ (degree 4)	Lesser / Greater GCT + octahedra; Unborn Axes + Greater GCT, Icosaxes + Lesser GCT (Main Theorem); Golden Cut Cubes + tetrahedra. Uniform: 3/2, 5/2, 2 + √2
24-cell × 24-cell (μ ∈ [1, 2])	√2 (self-dual point, μ ↔ 2/μ); φ, (15 + √5)/10; √5 − 1 = 2/φ on the dual side	bitruncated 24-cell, 48 truncated cubes (the C·O sweep at 1 + √2); Icosaxes / Unborn Axes cells beside cubes
600-cell × 120-cell (R = 1, $r^2 = (7 + 3\sqrt{5})/16$)	(13 + 5√5)/28, (111 + 45√5)/244; 2 − √5/2, (19 + 9√5)/44; 3(√5 − 1)/4, (5 + 2√5)/10	Lesser / Greater GCD + tetrahedra; Greater GCI + Lesser GCT, Lesser GCI + Greater GCT (bitruncated stage); Greater / Lesser GCT + icosahedra. Uniform: (15 + 8√5)/38, (11 + 3√5)/20, (9 + √5)/12

Table 2. Golden and notable stations of the three four-dimensional sweeps (s5, 17 checks). Values of μ exact; cell types and corner counts floating point except for the tesseract and 24-cell stations named in the text.

The three uniform values of the 600-cell × 120-cell family are the edge-equalising cuts of #76 (Proposition 1) [17], by Lemma 1 with R = 1 and the 600-cell's edge 1/φ (its dual side for the 120-cell): the truncated 600-cell, each edge cut 1/3 of the way from each end, at $\mu = 1 - (1/3)(1/\varphi^2)/2 = (9 + \sqrt{5})/12$; the bitruncated stage, #76's midpoint, whose corners lie 3/5 of the way from the middle of each triangle's edge to its centre, at $\mu = (11 + 3\sqrt{5})/20$, the scalar product of such a corner with either end of that edge; and the truncated 120-cell, each edge cut $(5 - \sqrt{5})/10$ of the way from each end, at $\mu = (15 + 8\sqrt{5})/38$ (exact, cross-links study). The DICTO All-Golden Polytopes of Proposition 7 lie in the bitruncated stage on either side of #76's midpoint, μ ≈ 0.8820 and 0.8892 against 0.8854.

Proposition 7 (DICTO All-Golden Polytopes in the 120-cell family; from Lemma 6). In the bitruncated stage of 600-cell × 120-cell, the Greater Golden Cut Icosahedron cells and the Lesser Golden Cut Tetrahedron cells occur at the same $\mu = 2 - \sqrt{5}/2$, and the Lesser Golden Cut Icosahedron cells and the Greater Golden Cut Tetrahedron cells at the same $\mu = (19 + 9\sqrt{5})/44$ (R = 1); these two stages are DICTO All-Golden Polytopes. Both values are exact consequences of the linear cell rule; the rule itself is checked exactly at three points only.

The reason is the one of Section 3: in the bitruncated stage the shared hexagons are truncated triangles of both cells, so a ratio φ on one side is 1/φ on the other. The 24-cell pair selects the silver ratio at its self-dual point $\mu = \sqrt{2}$, where the bitruncated 24-cell has 48 truncated cubes (exact); its Icosaxes and Unborn Axes cells sit at φ and $(15 + \sqrt{5})/10$ beside cubes.

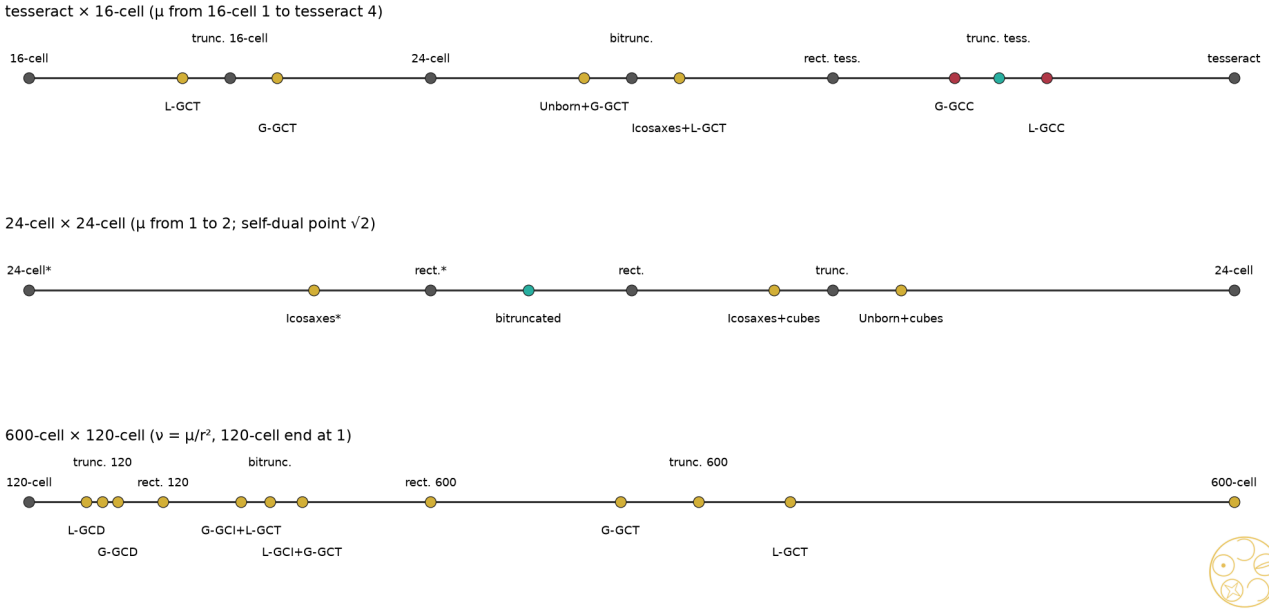


Figure 3. The three four-dimensional sweeps on their parameter lines, coloured by the field of μ as in Figure 1. Top: tesseract × 16-cell, μ from the 16-cell (1) to the tesseract (4), with the DICTO All-Golden Polytopes Unborn+G-GCT ($2 + 1/\varphi^2$) and Icosaxes+L-GCT (φ^2) either side of the bitruncated tesseract, and the Golden Cut Cubes (red) beside the truncated tesseract. Middle: 24-cell × 24-cell, μ from 1 to 2, the bitruncated 24-cell (teal) at the self-dual point √2. Bottom: 600-cell × 120-cell on $\nu = \mu/r^2$, the 120-cell end at 1, with the DICTO All-Golden Polytopes G-GCI+L-GCT and L-GCI+G-GCT at the bitruncated stage. G and L: Greater and Lesser; GCT, GCI, GCD, GCC: Golden Cut Tetrahedron, Icosahedron, Dodecahedron, Cube.

7. Cycles: nested chains closing on a scaled copy

A tight link A → B scales B down to the smallest copy that contains A, so that A touches B; the closure of a cycle is the product of its tight factors. **Two-link rule:** the cycle P ⊂ H(μ) ⊂ νP closes at ν, since P's corners are corners of H(μ) and the corners μn of μP* lie at ν on P's face normals; so every station's ν is a closure. A tight cycle that closes at a power of φ is one of the **DICTO Golden Tight Cycles**, DICTO's name for them.

pair	cycle (tight links)	closure
all	sweep cycle P* → L* → A* → G* → rectified → G → A → L → P → P*; each link but the last touches along P's faces	$(R/r)^2$: D·I $15 - 6\sqrt{5}$, C·O 3, T·T 9
D·I	D → icosidodecahedron → Greater Golden Cut Dodecahedron → triakis icosahedron $H(\mu_{TD})$ → D; factors $(9 - 3\sqrt{5})/2$, 1, $(16 + 7\sqrt{5})/33$, $2\sqrt{5} - 3$	φ
D·I	D → rhombic triacontahedron → icosidodecahedron → Lesser Golden Cut Icosahedron → D; factors 1, $3 - \varphi$, $1/2 + 3\sqrt{5}/10$, 1	φ
C·O	cube ⊂ H(φ) (the hull at Icosaxes) ⊂ φ·cube	φ
C·O	cube → octahedron → Icosaxes → truncated octahedron → H(φ) → cube	$\varphi^3 = 2 + \sqrt{5}$
C·O	cube ⊂ H(1 + √2) (triakis octahedron) ⊂ (1 + √2)·cube	$1 + \sqrt{2}$ (silver)
T·T	T → Lesser GCT (dual side) → Lesser GCT → truncated tetrahedron (dual side) → T	$\varphi^6 = 9 + 4\sqrt{5}$
Dharma pattern	P → rectified → Q(A*, ρ²) → H(A*) → P* → H(ρ²) → P (OG's analogue on each pair)	D·I $225/2 - 99\sqrt{5}/2$; C·O 9/2; T·T 27
120-cell (4D)	sweep cycle 600-cell* → … → 600-cell	$1/r^2 = 28 - 12\sqrt{5}$ (not golden)

Table 3. Tight cycles in D·I·C·T·O order of the pairs (s6, exact). L, A, G: Lesser golden, Archimedean, Greater golden stations; * the dual side.

The closure $(R/r)^2$ of the sweep cycle, $15 - 6\sqrt{5}$, 3 and 9, is the containment constant of #43, attained by the Platonic solids (Theorem 1), and for the icosahedral pair valid for every solid with icosahedral symmetry and sharp (Theorem 2) [18], and the two-step gain of DICTO's growth rule for the Platonic solids in #80 (Corollary) [19]. The same number governs #81's

core turn [20]: S(μ) has its faces at distance μ/R on the corner axes of P and r on its face axes, and by #81's Theorem (a) the core of a turn swaps the two distances, so it is similar to S(μ′) with $\nu' = (R/r)^2/\nu$, an involution of the open sweep. On C·O it is $\mu \mapsto 3/\mu$, which pairs the truncated octahedron (3/2) with the cuboctahedron (2) and fixes $\mu = \sqrt{3}$, where the two kinds of face are equally far from the centre; it differs from the space-filling partner $\mu \mapsto 3 - \mu$ of #90 (Proposition 6) [3], which pairs Icosaxes with Unborn Axes and fixes Kelvin's 3/2. On T·T it is the self-duality $\mu \leftrightarrow 1/\mu$ of Section 4.

The search covered all tight cycles of at most five links from P back to P among each pair's 17 or 18 stations, in floating point with every quoted cycle re-checked exactly (s6). With φ^k as the target it found 119 DICTO Golden Tight Cycles for D·I, all closing at φ; 73 + 165 + 4 for C·O, closing at φ, φ^2 and φ^3; and 255 for T·T, all at φ^6. None for D·I closes at φ^2 or φ^3. The criterion, every link tight, is that of #34 [21], where none of the 8,742 cycles through the Hecatodihedron, OG, the truncated icosahedron and the pentakis dodecahedron, with any of the icosahedron, icosidodecahedron, dodecahedron and rhombic triacontahedron, closes on a power of φ; the D·I cycles of Table 3, each through a golden-cut station, close at φ. The Dharma pattern closes golden on none of the three pairs, and replacing its OG station by the Lesser Golden OG leaves the closure unchanged.

Cycles with the rhombic dodecahedron and the truncated octahedron. The truncated octahedron (TO) is S(3/2) and the rhombic dodecahedron (RD) is H(2) = conv(cube ∪ 2·octahedron), the polar of the cuboctahedron (14 corners, 12 rhombi; checked). The cycle cube ⊂ RD ⊂ 2·TO ⊂ 2·cube has tight factors 1, 2, 1 and closure 2; cube → RD → triakis octahedron → TO → cube closes at $1 + \sqrt{2}$, and cube → RD → TO → H(φ) → cube at φ^2 (s8, exact). Counting cycles of at most five links that contain the RD or the TO: 436 close at 2, 320 at $1 + \sqrt{2}$, 109 at φ^2, 12 at $(1 + \sqrt{2})^2$ and 2 at φ^3 (floating-point search, examples exact). All 64 cycles from the cube through ordered subsets of the space-fillers {RD, TO, Icosaxes, Unborn Axes} are exact; their closures include 2, 3/φ (cube → Icosaxes → cube) and $(15 + 3\sqrt{5})/10$, and none is a power of φ. As nestings the space-fillers give rational or non-golden closures only; the space-filling cycles that do exist are the honeycomb slices of Section 8, where every stage is a filling.

8. Cycles as slices through hyper-lattices

A family of parallel slices of a periodic honeycomb repeats with the honeycomb's period, so the honeycomb itself is the cycle (s7, 14 checks, exact).

C·O, cube side. The rectified tesseractic honeycomb, sliced at $y_1 = h$, runs cube → Lesser GCC → truncated cube → Greater GCC → cuboctahedron and back, with period 2 (#89, reproduced) [2].

C·O, octahedron side. The 24-cell, $|x|_\infty \le 1$ and $|x|_1 \le 2$, is the Voronoi cell of $2Z^4 \cup (2Z^4 + (1, 1, 1, 1))$. Sliced cell-first at $y_1 = h$ its honeycomb gives the body-centred cubic pair C ∩ (2 − h)O at $2Z^3$ and C ∩ (1 + h)O at $2Z^3 + (1, 1, 1)$: cuboctahedra and octahedra at h = 0, **Icosaxes + Unborn Axes at h = 1/φ²**, two truncated octahedra (Kelvin's filling [34]) at h = 1/2, Unborn Axes + Icosaxes at h = 1/φ, and octahedra and cuboctahedra at h = 1; the volumes add to 8 per cell for every h, identically. This is #48's family of mixed truncated octahedra [8], now read as slices of one honeycomb. Sliced vertex-first (the D4 Voronoi cells [32] $|x_i| + |x_j| \le 1$ at $x_4 = h$), the same honeycomb gives rhombic dodecahedra on the face-

centred cubic lattice at h = 0, then RD ∩ (1 − h)·cube with small cubes on the other coset, then cubes at h = 1/2, the volumes adding to 2 per cell for every h.

> **Lemma 8 (slices of a Voronoi honeycomb; classical, Aurenhammer [22]).** The slice $x_4 = h$ of the Voronoi honeycomb of a point set in R^4 is the power diagram in R^3 of the sites $p = (p', p_4)$, projected to p', with weights $w = -(h - p_4)^2$.

Proof. For $x = (x', h)$, $|x - p|^2 = |x' - p'|^2 + (h - p_4)^2 = |x' - p'|^2 - w$, and the power cell of p' is the set where this is least. □ So both slices are lanes of the web of bonds of #52 and #53 [23, 24], with $\lambda = w_A - w_B$ as in #53. Cell-first, for $0 \le h \le 1$, only the sites at $x_4 = 0$ (A, on $2Z^3$, weight $-h^2$) and $x_4 = 1$ (B, on $2Z^3 + (1, 1, 1)$, weight $-(1 - h)^2$) can have cells: the caesium chloride sites in #53's units (half cube edge 1), at $\lambda = (1 - h)^2 - h^2 = 1 - 2h$. As h runs from 0 to 1, λ runs over #53's gradual hand-over interval $[-1, 1]$ from flip to flip, which is #48's golden loop, $\lambda = (x - 1)/(x + 1)$; Icosaxes and Unborn Axes at $h = 1/\varphi^2$ are $\lambda = \sqrt{5} - 2$, $x = \varphi$. Vertex-first, the D4 sites at $x_4 = 0$ and 1 are the two face-centred cubic lattices of rock salt (Na–Cl distance 1), again with $\lambda = 1 - 2h$: the rhombic dodecahedra at h = 0 are the end $\lambda = 1$ of #53's rock-salt lane, the cubes at h = 1/2 its switch point $\lambda = 0$, and its stop $\lambda = 4 - 2\sqrt{3}$, where every edge has one length (cells with 6 squares and 12 hexagons, beside cubes), is the slice at $h = \sqrt{3} - 3/2$ (Lemma 8 and the λ values checked in the cross-links study).

T·T. The tesseractic honeycomb sliced vertex-first has period 1; at $s \equiv 1/\varphi^2$ its tiles are the Lesser and the Greater Golden Cut Tetrahedra, with two small tetrahedra (exact).

D·I. There is no periodic version: by Theorem 3 every lattice with icosahedral symmetry has rank at least 6, and in Z^6 the symmetric flats are fixed, so a moving slice breaks the symmetry. What replaces the honeycomb is the finite 600-cell × 120-cell family of Section 6, where the D·I and T·T sweeps appear as cells.

Golden-tilted slice (a coding only). If the slice is tilted so that successive tile rows advance by 1/φ of the period, the tile heights are $\{k/\varphi\}$. Coded below or above the Unborn Axes height 1/φ they form the Fibonacci word (exact integer test, 300 letters), which links this family to DICTO's Fibonacci stackings: it is the word of #69's substitution c → ch, h → c, with the heights above 1/φ as the h layers, of share $1/\varphi^2$ [25]; #30 found the canonical Fibonacci chain along the spine of its five-fold chorus [26]. With four letters (p: cuboctahedron to Icosaxes, q: Icosaxes to Kelvin, r: Kelvin to Unborn Axes, s: Unborn Axes to octahedron) the word begins spsqpspsrpsqpsps. The tilted tiles are oblique sections, not the stations.

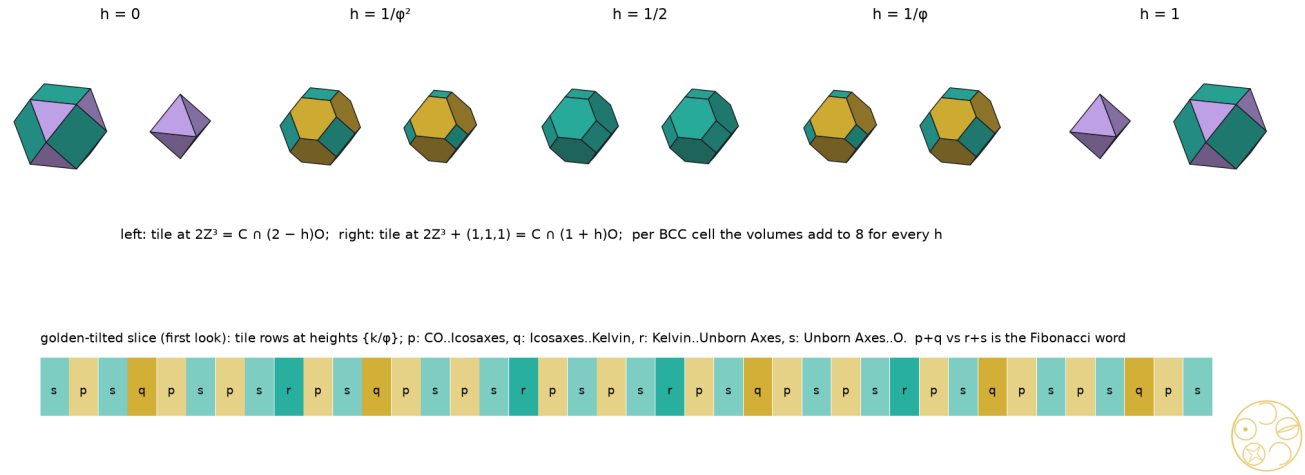


Figure 4. The 24-cell honeycomb sliced cell-first. Above, the two body-centred cubic tiles at h = 0, 1/φ², 1/2, 1/φ and 1: left the tile at $2Z^3$, C ∩ (2 − h)O; right the tile at $2Z^3$ + (1, 1, 1), C ∩ (1 + h)O; at 1/φ² they are Icosaxes and Unborn Axes, at 1/2 two truncated octahedra (Kelvin). Below, the first 40 letters of the four-letter word of a golden-tilted slice; p + q against r + s is the Fibonacci word.

Golden cut rhombic dodecahedra and triacontahedra as four-dimensional slices (floating point; negative within its scope). A slice with full O_h symmetry must be perpendicular to an axis whose stabiliser contains O_h: the coordinate axes for B4, the root directions for F4, and for I_h in H4 the 600-cell corner directions (by stabiliser orders). All 15 Wythoff polytopes [33] of each of B4, F4 and H4 were cut along these axes at 600 heights (H4: 150). The face signature of #89's golden cut rhombic dodecahedra (8 triangles, 6 squares, 12 octagons) occurs in 9 F4 stages and never in B4, but in none of them are the triangle and square sides equal where the octagon ratio could be φ: the only points with equal sides have ratio 0.2539. The signature of the golden cut rhombic triacontahedra (20 triangles, 12 pentagons, 30 octagons) occurs in 2 H4 stages, again never with equal corner sides (s9, s9b). Tiles of the uniform honeycombs are covered except sections of prism tiles away from their base. The trapezo-rhombic dodecahedron, the Voronoi cell of hexagonal close packing (DICTO's TRD node), has only D_{3h} symmetry; it is not a station of the cubic sweeps and was not reached by these slices.

9. Synthesis: golden and silver

pair	dim.	field	selected ratio, and why	golden stations (exact)
pentagon (c_5 = φ)	2	Q(√5)	φ native: the pentagon's diagonal	#88's rule: x = 1/3 and 1/(φ + 3)
square (c_4 = √2)	2	Q(√2)	silver: the diagonal √2; golden needs Q(√2, √5)	$x_G(4)$, $x_L(4)$, of degree 4
D·I	3 → 6	Q(√5)	φ native in the solids, but no lattice (Theorem 3); natural values ρ² = φ², μ = √5	all four on the sweep; OG at (μ_{TI}, φ²); Lesser and Greater Golden OG at (√5, φ²) and ((13 + 6√5)/11, φ²)
C·O	3 → 6	Q(√2) and Q(√5)	octahedron side φ (Icosaxes, roots at μ = φ); cube side silver 1 + √2	Icosaxes φ, Unborn Axes 3 − φ; Golden Cut Cubes of degree 4
T·T	3 → 6	Q(√5)	golden in the tesseract's height (h = 1/φ², 1/φ); no lattice selection (norms 34, 46)	2√5 − 3, 1 + 2/√5 and their reciprocals
tesseract × 16-cell	4 → 8	Q, Q(√5), Q(√2, √5)	golden heights in the two lower unit layers; √2 in the top layer	φ², 2 + 1/φ² (DICTO All-Golden Polytopes), φ, 3 − φ
24-cell × 24-cell	4 → 8	Q(√2)	silver at the self-dual point √2 (bitruncated 24-cell)	φ, (15 + √5)/10 (Icosaxes / Unborn Axes cells)
600-cell × 120-cell	4 → 8	Q(√5)	φ native; golden cells at every stage; linear cell rule (Lemma 6)	2 − √5/2, (19 + 9√5)/44 (DICTO All-Golden Polytopes)

Table 4. The pattern across dimensions. The pentagon's golden cuts follow from #88's c_n rule.

In two to four dimensions the five-fold structures are native: the pentagon, the D·I pair and the 120-cell and 600-cell, all in Q(√5). Their golden stations are plentiful but are never picked out by a lattice, because no lattice carries their symmetry below dimension 6. In six and eight dimensions the golden ratio appears hidden in a cubic lattice: the Fibonacci slope of $Z^3 \oplus Z^3$ selects Icosaxes [3], and the tesseract's and 24-cell's golden heights carry golden cells. The cube side is the exception throughout: there the square's √2 wins, giving the silver ratio 1 + √2 as the truncated cube's μ, the 24-cell's self-dual point and a silver two-link cycle.

10. Prior art and what is new

Archimedean solids as the common part of a Platonic solid and its dual go back to Kepler [27]; truncation at any depth, and Conway's ambo and truncation operations, are standard [28]; polarity and the common part $P \cap \mu P^*$ are those of convex-polytope theory [29]. The bitruncated tesseract as the common part of the tesseract–16-cell compound and the 24-cell's bitruncation halfway to its dual are classical [30, 31], as are sections of the regular four-dimensional solids (Boole Stott [9]; Coxeter [10]) and the 24-cell honeycomb's slices, vertex-first (rhombic dodecahedral and cubic honeycombs) and cell-first (rectified and bitruncated cubic honeycombs). Zefiro studied truncations of cubic solids by the rhombic dodecahedron and the cube [12]. The icosahedral cut-and-project of Z^6 is due to Kramer and Neri, Duneau and Katz, and Elser [13–15], and icosahedral polyhedra from D6 to Al-Siyabi, Koca and Koca [11]. #48 gave the mixed truncated octahedra [8]; #88, #89 and #90 precede this paper [1–3]; the polar Dharma cycle is #32's [6]. The closure $(R/r)^2$ is #43's constant and #80's gain [18, 19], the core turn is #81's [20], the polar side of C·O is #54's Theorem B [16], the uniform four-dimensional stations are #76's [17], the honeycomb slices are lanes of #52 and #53 [23, 24], and the tight-cycle criterion is #34's [21]. The coding of irrational rotations by Sturmian and Fibonacci words is classical.

New within these checks. Lemma 1 as a statement about all corner cuts of a dual pair; the DICTO All-Golden Polytopes of the Main Theorem and Proposition 7; the golden stations of the D·I and T·T sweeps as sections of the six-dimensional products, with Theorems 2–4; OG as $Q(\mu_{TI}, \varphi^2)$, the polar Dharma cycle as stations of the D·I families, and the Golden OGs; the DICTO Golden Tight Cycles; the 24-cell honeycomb slices as a cycle holding Icosaxes and Unborn Axes. Where an earlier source has one of these solids, it stays in DICTO's family, found independently, and the source is credited.

Sources checked (3 November 2026). Reference library (kaleidohedra-references; reading list searched for truncation, Wythoff, Kelvin, bitruncated, Fibonacci) and a light web search (Archimedean solids as the intersection of a Platonic solid and its dual, e.g. G. W. Hart's pages; Wikipedia, “Truncated tesseract”, “Truncated 24-cells”, “24-cell honeycomb”). Not found: the golden stations of the D·I and T·T sweeps as sections of six-dimensional products, OG as $Q(\mu_{TI}, \varphi^2)$, the DICTO All-Golden Polytopes, or the DICTO Golden Tight Cycles. Boole Stott and Coxeter are cited for their well-known contents and were not re-read for this paper. The search is not exhaustive.

11. Conclusion

Every corner cut of a regular solid is a slice of the solid's product with its dual, so the golden cuts of #88 are stations of three sweeps. One dimension up the stations meet: in the tesseract trimmed by the 16-cell the golden heights of the cube cells and the tetrahedral cells coincide,

and two four-dimensional solids, the DICTO All-Golden Polytopes of the tesseract, have nothing but golden cells, Icosaxes with the Lesser Golden Cut Tetrahedron and Unborn Axes with the Greater. The dodecahedron's pair lives wholly in Q(√5); it carries DICTO's OG, his Lesser and Greater Golden OG and the polar Dharma cycle as sections and hulls of one six-dimensional product, but no lattice carries its symmetry. DICTO Golden Tight Cycles close at φ, honeycomb slices give the golden space-filling pair of #48, and wherever the square enters, √2 turns golden into silver.

> **Data and code.** studies/2026-11-03-dual-pairs-dimensions: dp_lib.py (on sixd_lib.py and cubes_lib.py of the #90 study, unchanged); s1_sweeps.py (Lemma 1, Theorem 2, Table 1; co/di/tt: 22, 20, 20 checks), s2_di_lattice.py (Theorem 3; 9), s3_tt.py (Theorem 4; 9), s4_og_dharma.py (Theorem 5; 21), s4b_og_lattice.py (lifts; 3), s4c_relatives.py (Golden OGs; 2 + 2), s5_fourd.py (Main Theorem (b), Lemma 6, Proposition 7, Table 2; 17), s6_cycles.py (Table 3 and the search; co/di/tt: 10, 12, 10), s7_hyper.py (Section 8; 14), s8_rd_to.py (RD and TO cycles; 4), s9_golden_rhombic.py and s9b_rhombic_ratios.py (B4, F4, H4 search, floating point), figs.py (figures). All checks pass; s5 asserts the corner set of the 16-cell section (signed permutations of (1, …, 1, t, 0, …)) and the exact derivation from the 24 inequalities was done in the referee check, while the floating-point section derives it. Run with Python 3.

Open points. Lemma 6 is not proved in general (#76's exact constructions of the uniform stages could make it exact at its three stations); the four-dimensional search for golden cut rhombic solids leaves out sections of prism tiles away from their base; the golden-tilted slice of Section 8 is a coding of tile heights, not a list of exact stations; the trapezo-rhombic dodecahedron (hexagonal close packing) lies outside the cubic sweep; and the factor 2 between the OG lift scale 342 here and 171 in the bridge study has not been re-derived.

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